Fiscal Year:	FY 2005	**Task Last Updated:**	FY 05/24/2005
PI Name:	Troian, Sandra M Ph.D.		
Project Title:	MICROFLUIDS: ACTUATION BY MODULATION OF SURFACE STRESSES-FLOW MODELING TO DEVICE IMPLEMENTATION		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	01/28/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Darhuber, Anton (Princeton University)		
Grant/Contract No.:	NNC04GA52G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The project has experimental and theoretical components. Theory will be used to guide design. Variations of the small-slope theory for free surface flows will be used taking advantage of the small height compared to the planar extension of the fluid bodies involved. To develop an innovative microfluidic chip based on an open and electronically addressable format for actuation and detection of microliter liquid samples. The invention of the integrated circuit in the 1950's and large-scale electronic integration in the 1970's revolutionized the US economy by introducing microscale circuitry controlling everything from household appliances to rocket propulsion. A similar revolution is taking place in the field of microfluidics, where the functions of once heavy and expensive analytic and diagnostic instruments are being miniaturized for use in portable and low power units. Devices based on pneumatic pumping have shown, for example, how multiplexed addressability and control allow fluidic sampling of thousands of channels with arbitrary complexity. Microfluidic devices take full advantage of the technology developed for ICs and micro-electro-mechanical		

systems (MEMS) as well as the simple, rapid fabrication techniques based on elastomeric materials like silicones. Microfluidic devices show enormous promise for genomic research, drug design, light and flexible display technologies, tunable fiberoptic waveguides, clinical and forensic analysis, advanced environmental monitoring for extended human space flight, biochemical agent detection and even soil remediation. For purposes of space travel, miniaturization of on-board clinical testing kits could dramatically decrease the cost, sample volume, time and weight required by current instrumentation. Microfluidic devices also offer tremendous opportunities for fundamental research in fluid physics at the micron and nanoscale.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

• Flow Modeling and Experimentation on Mixing
Mixing of liquid samples at low Reynolds number poses special difficulty. Actuation mechanisms suitable for free surface flows like thermocapillary forcing raise a number of interesting questions involving efficient mixing configurations especially in systems with exceedingly large surface to volume ratio. We have investigated the characteristics of convective and diffusive mixing in continuous-mode streaming of thermocapillary flows on chemically micropatterned surfaces. Mixing times and mixing lengths relevant to chemical microreactors and gas sensors are investigated for various geometries and parameter ranges. Using both scaling arguments and full numerical solutions, we have extracted optimal operating conditions. Our experiments for monitoring the interfacial diffusive broadening in adjacent flowing microstreams based on confocal fluorescence microscopy confirm the numerical predictions. Three important mixing regimes, based on analogues of purely diffusive dynamics, Rhines-Young shear-augmented diffusion and Taylor-Aris dispersion, have been identified and investigated for use in thermocapillary devices.

• Device Implementation

Electronic monitoring and control of arbitrary surface temperature distributions is a necessary requirement for the development of a miniaturized and electronically addressable fluidic chip based on thermocapillary actuation. We have designed a microfluidic chip suitable for controlling the position and speed of liquid droplets (or continuous streams) on a glass surface by means of embedded microheater arrays. The microheaters provide control over the surface temperature distribution with high spatial resolution. The desired temperature gradients, and therefore the local thermocapillary stress imposed on liquid samples, is electronically programmable. In combination with substrates of mixed wettability achieved through self-assembling monolayers, this device is being explored for use as a logistic platform for parallel and automated routing, splitting, mixing and reacting of a multitude of liquid samples. We have investigated droplet motion and splitting using alkanes, poly(ethylene glycol) and water.

Book Chapters:

-General Approach to the Linear Stability of Thin Spreading Films, J. M. Davis, B. J. Fischer and S. M. Troian in Interfacial Fluid Dynamics and Transport Properties , R. Narayanan and D. Schwabe, Eds., Lecture Notes in Physics Series, pp. 79-105 (Springer-Verlag, Berlin 2003).

-A Study of Mixing in Thermocapillary Flows on Micropatterned Surfaces, A.A. Darhuber, J. Z. Chen, J. M. Davis and S. M. Troian, in "Transport and Mixing at the Microscale", J. M. Ottino and S. Wiggins, Eds., Phil. Trans. Royal Soc. London A, Vol. 362, No. 1818, pp. 1037-1058 (2004).

-Principles of microfluidic actuation by modulation of surface stresses, A.A. Darhuber and S.M. Troian, Ann. Rev. Fluid Mech., Vol. 37 (2005), in press.

Refereed Publications

-Capacitive Sensing of Droplets for Microfluidic Devices Based on Thermocapillary Actuation, J.Z. Chen, A. A. Darhuber, S. M. Troian and S. Wagner, Lab on a Chip 4, 473-480 (2004)

-Influence of Boundary Slip on the Optimal Excitations in Thermocapillary Driven Spreading, Jeffrey M. Davis and Sandra M. Troian, Phys. Rev. E (2004), in press.

-Effect of Contact Angle Hysteresis on Thermocapillary Droplet Actuation, Jian Z. Chen, Sandra M. Troian, Anton A. Darhuber and Sigurd Wagner, in press for J. Appl. Phys.

-Layering Transitions and Squeeze-Out Patterns in Nanometer Polymeric Film, Steffen Berg and Sandra M. Troian, in review for Science.

-Images of the Floating World: Drainage Patterns in Thinning Soap Films, Steffen Berg, Eric A. Adelizzi and Sandra M. Troian, Phys. Fluids 16 (9), S6 (2004).

Below is a link to our winning entry for the 2004 Gallery of Fluid Motion involving thin film instabilities driven by Marangoni effects in microscale films containing associating polymer-surfactant solution. Images and video sequence can be viewed at http://pof.aip.org/ (Ed. note 3/20/2017--URL no longer available but see http://www.troian.caltech.edu/ for their entries). The September 2004 issue of Phys. Fluids contains a limited set of images with a description of experimental details.

Bibliography Type: Description: (Last Updated: 03/26/2021)

Articles in Peer-reviewed Journals

Chen JZ, Darhuber AA, Troian SM, Wagner S. "Capacitive sensing of droplets for microfluidic devices based on thermocapillary actuation." Lab Chip. 2004 Oct;4(5):473-80. Epub 2004 Jun 25. https://doi.org/10.1039/b315815b ; PMID: 15472731 , Oct-2004

Articles in Peer-reviewed Journals	Darhuber AA, Chen JZ, Davis JM, Troian SM. "A study of mixing in thermocapillary flows on micropatterned surfaces." Philos Trans A Math Phys Eng Sci. 2004 May 15;362(1818):1037-58. https://doi.org/10.1098/rsta.2003.1361 ; PMID: 15306483 , May-2004
Articles in Peer-reviewed Journals	Darhuber AA, Troian SM. "Principles of microfluidic actuation by modulation of surface stresses." Annu Rev Fluid Mech. 2005;37:425-55. https://doi.org/10.1146/annurev.fluid.36.050802.122052 , Jan-2005
Articles in Peer-reviewed Journals	Berg S, Adelizzi EA, Troian SM. "Images of the floating world: Drainage patterns in thinning soap films." Phys Fluids. 2004 Sep;16(9):S6. https://doi.org/10.1063/1.1763921 , Sep-2004
Books/Book Chapters	Davis JM, Fischer BJ, Troian SM. "General Approach to the Linear Stability of Thin Spreading Films." in "Interfacial Fluid Dynamics and Transport Properties, Lecture Notes in Physics Series." Ed. R. Narayanan, D. Schwabe. Berlin : Springer, 2003. p. 79-106. https://doi.org/10.1007/978-3-540-45095-5_5 , Jan-2003